its marginals, obtained by integrating out two variables, e.g. $W_{|\xi\rangle_2}(x_A, x_B) = \int W_{|\xi\rangle_2}(x_A, p_A, x_B, p_B) dp_A dp_B$. Plots of some marginals are shown in Fig. 3.6.

It is clear that two-mode squeezed states display correlations between the quadratures of modes A and B. Specifically, $\hat{p}_A(\theta/2)$ and $\hat{p}_B(\theta/2)$ will be correlated while $\hat{x}_A(\theta/2)$ and $\hat{x}_B(\theta/2)$ will be anti-correlated, as can be seen in Fig. 3.6 for $\theta = 0$. In the limit of infinite squeezing ($r \to \infty$), the correlations become perfect, i.e. $\hat{p}_A(\theta/2) = \hat{p}_B(\theta/2)$ and $\hat{x}_A(\theta/2) = -\hat{x}_B(\theta/2)$. These correlations are quantum in nature and in fact the two-mode squeezed states are entangled.

In the Fock basis, the two-mode squeezed vacuum state exhibit perfect correlation between the excitation numbers in the two modes. It can be expressed

$$|\xi\rangle_2 = \operatorname{sech}(r) \sum_{n=0}^{\infty} (-1)^n e^{i\theta n} \tanh^n(r) |n, n\rangle_{AB}. \tag{3.120}$$

We see that the excitation numbers in modes A and B are equal in all terms. For low squeezing ($r \ll 1$),

$$|\xi\rangle_2 \approx \frac{1}{\sqrt{1+r^2}} \left(|0,0\rangle - e^{i\theta} r |1,1\rangle \right), \tag{3.121}$$

where we have dropped terms of order $O(r^2)$ or higher. In this limit, $|\xi\rangle_2$ can thus be understood as a partially entangled state of two qubits. In the opposite limit of infinite squeezing ($r \to \infty$), $\tanh(r)$ tends to 1 and thus $|\xi\rangle_2$ tends towards an equally weighted superposition of all excitation numbers. Such a superposition is not normalisable – the limit of infinite squeezing is not physical since it would require infinite energy. However, considering maximally entangled states in dimension d, such as

$$\frac{1}{\sqrt{d}} \left(|0,0\rangle + |1,1\rangle + \cdots + |d-1, d-1\rangle \right), \tag{3.122}$$

the limit of infinite squeezing for $|\xi\rangle_2$ can be understood as an infinite-dimensional maximally entangled state.

3.1.9 Cat states

Coherent states, single- and to-mode squeezed states, as well as the thermal states discussed in the next section all belong to the set of Gaussian states. ...

An interesting example of non-Gaussian states are superpositions of coherent states. Coherent-state superpositions are often known as *cat states* in analogy with the famous thought experiment of Schrödinger because, for large amplitudes, different coherent states are macroscopically distinct (and coherent states are also in a sense the 'most classical' quantum states of light, as discussed in Sec. 3.1.5). In particular, we can define the so-called even and odd cat states

$$|C_{\pm}^{\alpha}\rangle = \frac{1}{N_{\pm}} (|\alpha\rangle \pm |-\alpha\rangle), \tag{3.123}$$

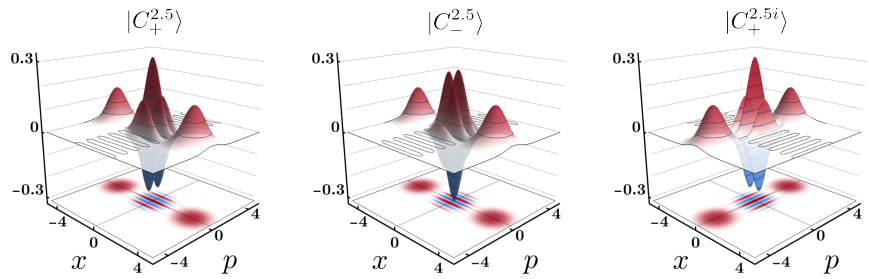

Figure 3.7: Wigner functions for even (left) and odd (middle) cat states with real amplitude $\alpha = 2.5$, and an even cat state with complex $\alpha = 2.5i$ (right). Note how the interference pattern changes with the relative phase.

where $N_\pm = \sqrt{2 \pm 2\,\mathrm{Re}\langle\alpha|-\alpha\rangle}$. In the Fock basis, from (3.66) we have

$$|C_+^\alpha\rangle = \frac{2e^{-\frac{1}{2}|\alpha|^2}}{N_+}\left(|0\rangle + \frac{\alpha^2}{\sqrt{2!}}|2\rangle + \frac{\alpha^4}{\sqrt{4!}}|4\rangle + \cdots\right), \tag{3.124}$$

$$|C_-^\alpha\rangle = \frac{2e^{-\frac{1}{2}|\alpha|^2}}{N_-}\left(\alpha|1\rangle + \frac{\alpha^3}{\sqrt{2!}}|3\rangle + \frac{\alpha^5}{\sqrt{5!}}|5\rangle + \cdots\right). \tag{3.125}$$

It should be clear why $|C_+^\alpha\rangle$ is called the even cat – it has support only on even Fock states. Similarly, the odd cat $|C_-^\alpha\rangle$ has support only on odd Fock states. From this it also follows that the two states are orthogonal $\langle C_+^\alpha|C_-^\alpha\rangle = 0$.

The Wigner functions of the even and odd cat states can be expressed in terms of coherent-state Wigner functions as

$$W_{|C_\pm^\alpha\rangle}(x,p) = \frac{1}{N_\pm}\Big(W_{|\alpha\rangle}(x,p) + W_{|-\alpha\rangle}(x,p) \pm 2W_{|0\rangle}(x,p)\cos\Big(2\sqrt{2}\,[x\,\mathrm{Im}(\alpha) - p\,\mathrm{Re}(\alpha)]\Big)\Big). \tag{3.126}$$

Some examples are plotted in Fig. 3.7. Note that these functions are clearly not Gaussian. In addition to coherent components corresponding to the two coherent states in the superpositions, they feature an interference pattern that depends on the relative phase in the superpositions and includes regions where they become negative.

Cat states have been extensively studied both for their conceptual appeal and because they can be used to define error-correcting codes for protecting quantum information encoded into bosonic modes. Codes may use the even and odd cat or more general coherent-state superpositions, e.g. with four components with pases ± 1 and $\pm i$ (sometimes known as 4-headed cats). Cat states are potentially very useful, however, they are also generally challenging to produce, especially in optical systems. As they are non-Gaussian, they cannot be produced with linear optical operations alone starting from Gaussian states. Some nonlinear operation or measurement is required. In optical systems, the largest cat states so far have $\alpha < 3$, while in microwave cavities cat states with mean photon numbers ($\approx |\alpha|^2$) above 100 can be generated.